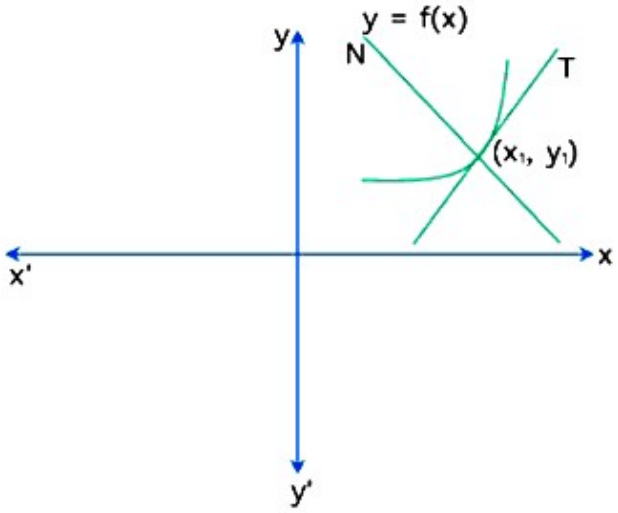
Application of Derivatives

DISCLAIMER

"The content provided herein are created and owned by various authors and licensed to Sorting Hat Technologies Private Limited ("Company"). The Company disclaims all rights and liabilities in relation to the content. The author of the content shall be solely responsible towards, without limitation, any claims, liabilities, damages or suits which may arise with respect to the same."

Application of Derivatives

Application of Derivatives (Tangent and Normal) :



$y = f(x)$ is given, then

$$\left.\frac{dy}{dx}\right|_{(x_1,y_1)} = \text{slope of tangent} = m$$

Equation of tangent :

$$\boxed{(y - y_1) = m(x - x_1)}$$

Equation of normal :

Slope of normal $m_N = -\frac{1}{m} = -\frac{1}{\left(\frac{dy}{dx}\right)_{(x_1,y_1)}}$

Equation: $y - y_1 = m_N (x - x_1)$

$$(y - y_1) = \frac{-1}{\left.\frac{dy}{dx}\right|_{(x_1,y_1)}}(x - x_1)$$

$$\boxed{(y - y_1) = -\left(\frac{dx}{dy}\right)_{(x_1,y_1)}(x - x_1)}$$

If the curve is expressed parametrically, then $x = \phi(t)$ and $y = \psi(t)$

$$m = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\psi'(t)}{\phi'(t)}$$

Equation of tangent :

$$\boxed{\{y - \psi(t)\} = \frac{\psi'(t)}{\phi'(t)}(x - \phi(t))}$$

Know the facts

Do check the point (x_1, y_1) lies on the curve $y = f(x)$

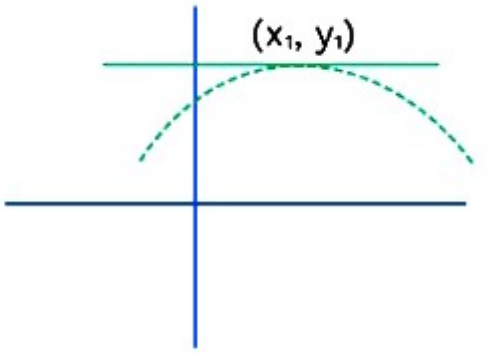
Equation of normal :

$$y - \psi(t) = -\frac{\phi'(t)}{\psi'(t)}(x - \phi(t))$$

Important points :

(1) If the tangent to the curve $y = f(x)$ at the point (x_1, y_1) is parallel to the x-axis, then $\left.\frac{dy}{dx}\right|_{(x_1,y_1)} = 0$ and if tangent is parallel to the line $ax + by + c = 0$

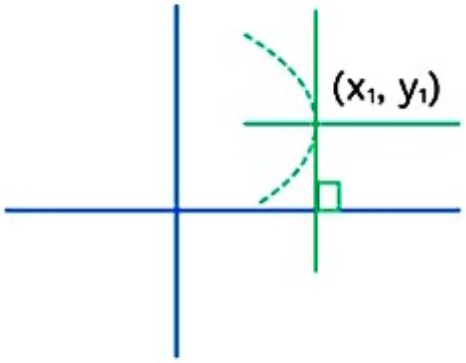
$$\Rightarrow \left.\frac{dy}{dx}\right|_{(x_1,y_1)} = \frac{-a}{b}$$



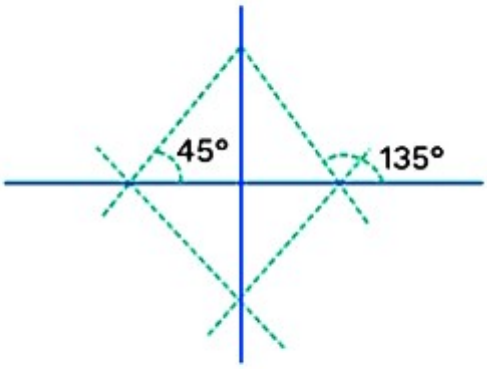
(2) If the tangent to the curve at (x_1, y_1) is perpendicular to x-axis then $\left.\frac{dx}{dy}\right|_{(x_1,y_1)} = 0$

and if it is perpendicular to the line $ax + by + c = 0$ then $\left.\frac{dy}{dx}\right|_{(x_1,y_1)} \times \left(\frac{-a}{b}\right) = -1$

assuming both slopes are finite.



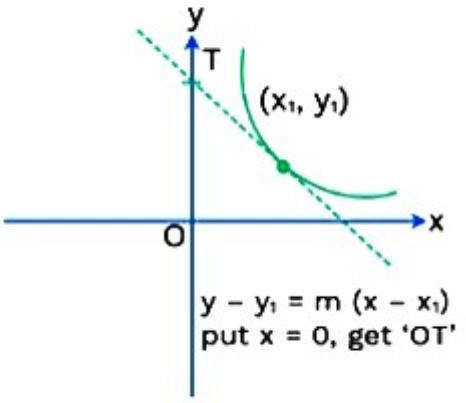
(3) If the tangent to the curve at (x_1, y_1) is equally inclined with the axis then $m = \pm 1$



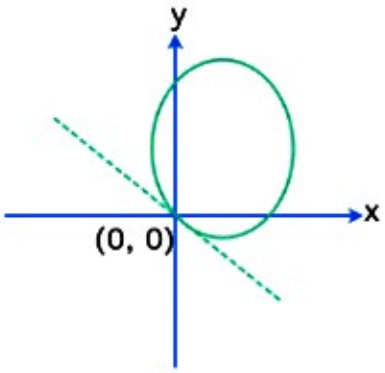
(4) If the tangent makes equal non-zero intercepts on coordinate axis then **m = -1**.

(5) If the tangent to the curve at (x_1, y_1) cuts off equal non-zero distances from origin on the coordinates axis, then **m = ± 1**.

(6) Initial ordinate of a tangent is y-intercept of tangent
OT(initial ordinate) = $y_1 - mx_1$

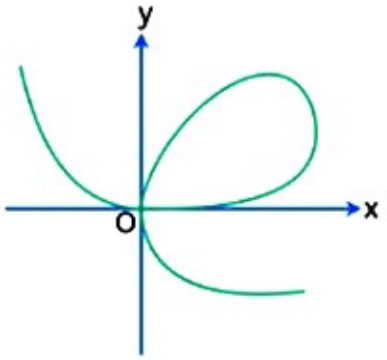


(7) If a curve passes through origin then equation of tangent to the curve at origin can be directly written by equating to zero the lowest degree term appearing in the equation of curve.



E.g. : Equation of circle passing through origin
$x^2 + y^2 + 2gx + 2fy = 0$
then equation of tangent to the circle at origin is $2gx + 2fy = 0$
$\Rightarrow gx + fy = 0$

E.g. : $x^3 + y^3 - 3xy = 0$ (folium of descartes)
Equation of tangent to the curve at origin is $x\,y = 0$. In this example, the same line is tangent as well as normal to the same curve at the same point.



(8) Parametric equation:

(i) $x^2 + y^2 = a^2 \quad \Rightarrow \quad x = a\cos\theta,\ y = a\sin\theta$

(ii) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1 \quad \Rightarrow \quad x = a\cos\theta,\ y = b\sin\theta$

(iii) $\sqrt{x} + \sqrt{y} = \sqrt{a} \quad \Rightarrow \quad x = a\cos^4\theta,\ y = a\sin^4\theta$

(iv) $x^{2/3} + y^{2/3} = c^{2/3} \quad \Rightarrow \quad x = c\cos^3\theta,\ y = c\sin^3\theta$

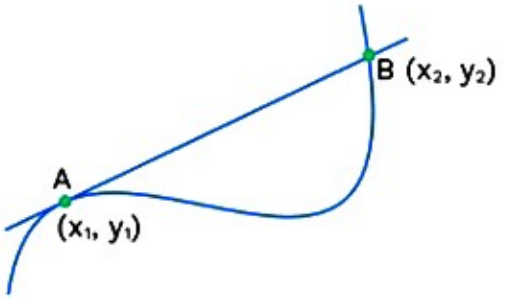
(v) $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1 \quad \Rightarrow \quad x = a\sec\theta,\ y = b\tan\theta$

(vi) $y^2 = 4ax \quad \Rightarrow \quad x = at^2,\ y = 2at$

(vii) $y^2 = x^3 \quad \Rightarrow \quad x = t^2,\ y = t^3$

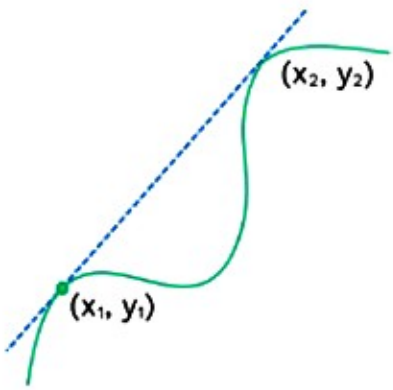
(9) Tangent to the curve at (x_1, y_1) intersect the curve again at (x_2, y_2).

$$\left.\frac{dy}{dx}\right|_{(x_1,y_1)} = \frac{y_2 - y_1}{x_2 - x_1}$$



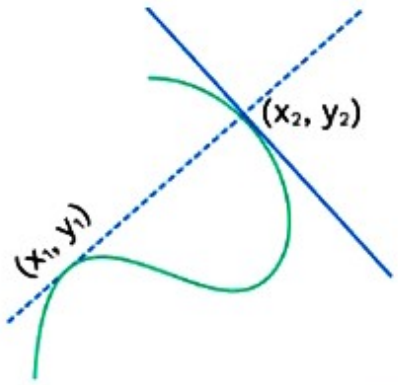
(10) Tangent to the curve at (x_1, y_1) is again tangent to the curve at same other point (x_2, y_2)

$$\left.\frac{dy}{dx}\right|_{(x_1,y_1)} = \left.\frac{dy}{dx}\right|_{(x_2,y_2)} = \frac{y_2 - y_1}{x_2 - x_1}$$



(11) Tangent to the curve at (x_1, y_1) again intersect the curve at (x_2, y_2) as normal.

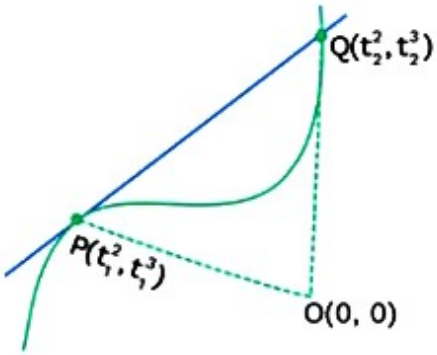
$$\left.\frac{dy}{dx}\right|_{(x_1,y_1)} = \frac{-1}{\left.\frac{dy}{dx}\right|_{(x_2,y_2)}} = \frac{y_2 - y_1}{x_2 - x_1}$$



Q.1 **Tangent to the curve $y^2 = x^3$ at the point P intersects the curve again at Q, then the value of $\frac{m_{OP}}{m_{OQ}}$ is (where O is the origin)**

(A) 1 **(B) –1** **(C) – 2** **(D) 1 or –2**

Sol. **(C)**



$$\frac{m_{OP}}{m_{OQ}} = \frac{t_1}{t_2}$$

$x = t^2, y = t^3$

$$\frac{dx}{dt} = 2t, \frac{dy}{dt} = 3t^2$$

$$\left.\frac{dy}{dx}\right|_{(t_1)} = \frac{3}{2}t_1 = m = \frac{t_2^3 - t_1^3}{t_2^2 - t_1^2} = \frac{t_2^2 + t_1^2 + t_1t_2}{t_1 + t_2}$$

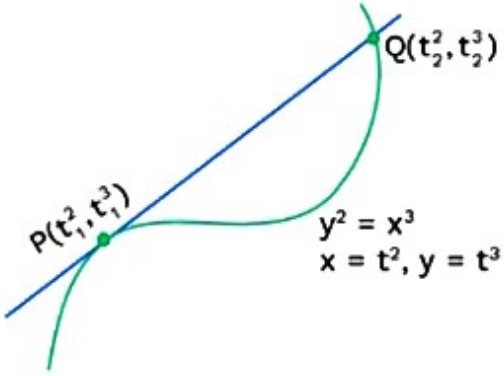
$\Rightarrow \quad 3t_1^2 + 3t_1t_2 = 2t_1^2 + 2t_2^2 + 2t_1t_2$

$\Rightarrow \quad 2t_2^2 - t_1^2 - t_1t_2 = 0$

$\Rightarrow \quad (t_2 - t_1)(2t_2 + t_1) = 0$

$\Rightarrow \quad 2t_2 = -t_1 \Rightarrow \frac{t_1}{t_2} = -2$

So $\frac{m_{OP}}{m_{OQ}} = -2$



Alternate Method :

Let the equation of tangent be $y = mx + c$

We get, $t^3 - mt^2 - c = 0$ $\quad t_1, t_1, t_2$

Sum of roots two at a time;

$t_1^2 + t_1t_2 + t_1t_2 = 0$

$\Rightarrow \quad t_1^2 + 2t_1t_2 = 0$

$\Rightarrow \quad t_1(t_1 + 2t_2) = 0$

$\Rightarrow$ either $t_1 = 0$ (not possible) or $t_1 = -2t_2$

$\Rightarrow \quad \frac{t_1}{t_2} = -2$

So, $\frac{m_{OP}}{m_{OQ}} = -2$

Q.2 **Equation of normal to the curve $x^2 = 4y$ which passes through (1, 2) is -**

(A) $2x + y = 4$ (B) $x + 2y = 4$ (C) $x + y = 3$ (D) None

Sol. **(C)**

Parametric point $(2t, t^2)$

$$\left.\frac{dy}{dx}\right|_{(t)} = \frac{x}{2} = \frac{2t}{2} = t$$

also slope of normal = $\frac{t^2 - 2}{2t - 1}$ and $-\frac{1}{t}$

which gives $\frac{t^2 - 2}{2t - 1} = -\frac{1}{t} \Rightarrow t(t^2 - 2) = 1 - 2t$

$\Rightarrow \quad t^3 - 2t = 1 - 2t$

$\Rightarrow \quad t^3 = 1 \Rightarrow t = 1$

for t = 1, point (2, 1)

So equation of normal,

$y - 1 = -1(x - 2)$

$y - 1 = -x + 2$

$x + y = 3$

Q.3 **A curve is defined parametrically as $x = e^{2t} + 2e^{-t}$ and $y = e^{2t} + e^t$. The equation of the tangent to the curve at $t = \ln 2$ is**

(A) $5x - 6y = 7$ (B) $5x - 3y = 7$ (C) $10x - 7y = 8$ (D) $3x - 2y = 3$

Sol. **(C)**

At $t = \ln 2$

$x = e^{2\ln 2} + 2e^{-\ln 2}$ and $y = e^{2\ln 2} + e^{\ln 2}$

$x = 4 + \frac{2}{2}$ and $y = 4 + 2$

$x = 5$ and $y = 6$

$$\frac{dx}{dt} = 2e^{2t} - 2e^{-t} \text{ and } \frac{dy}{dt} = 2e^{2t} + e^{t}$$

$$\left.\frac{dy}{dx}\right|_{t=\ell n2} = \left.\frac{2e^{2t} + e^{t}}{2e^{2t} - 2e^{-t}}\right|_{t=\ln 2} = \frac{2\times 4 + 2}{2\times 4 - \frac{2}{2}} = \frac{10}{7}$$

Equation of tangent: $y - 6 = \frac{10}{7}(x - 5)$

$\Rightarrow \quad 7y - 42 = 10x - 50$

$\Rightarrow \quad 10x - 7y = 8$

Q.4 **Prove that all the points on curve $y^2 = 4a\left(x + a\sin\left(\frac{x}{a}\right)\right)$ at which tangent is parallel to x-axis, lies on a parabola.**

Sol. Let the tangent to the parabola at (h, k) is parallel to x axis.

then $2y\frac{dy}{dx} = 4a\left(1 + \frac{a}{a}\cos\left(\frac{x}{a}\right)\right) = 4a\left(1 + \cos\left(\frac{x}{a}\right)\right)$

$0 = 1 + \cos\left(\frac{h}{a}\right) \Rightarrow \cos\left(\frac{h}{a}\right) = -1$

$\frac{h}{a} = (2n + 1)\pi$, where $n \in I$.

also $\quad k^2 = 4a\left(h + a\sin\left(\frac{h}{a}\right)\right)$

$\Rightarrow \quad k^2 = 4ah$

locus $\Rightarrow \quad y^2 = 4ax$ which is a parabola. **Hence proved.**

Q.5 **Tangents are drawn from the origin to the curve y = sin x. Prove that their point of contacts lie on the curve $x^2y^2 = x^2 - y^2$.**

Sol. (h, k) lies on the curve.

So $k = \sin h$ ...(i)

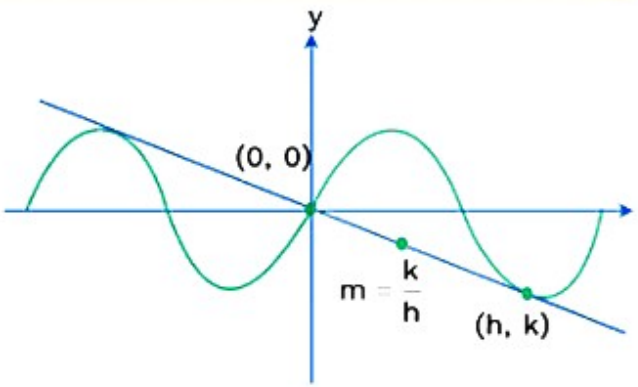
Slope: $\frac{dy}{dx} = \cosh = \frac{k}{h}$ ...(ii)

Squaring and adding,

$k^2 + \frac{k^2}{h^2} = 1$

$h^2k^2 = h^2 - k^2$

$x^2y^2 = x^2 - y^2$



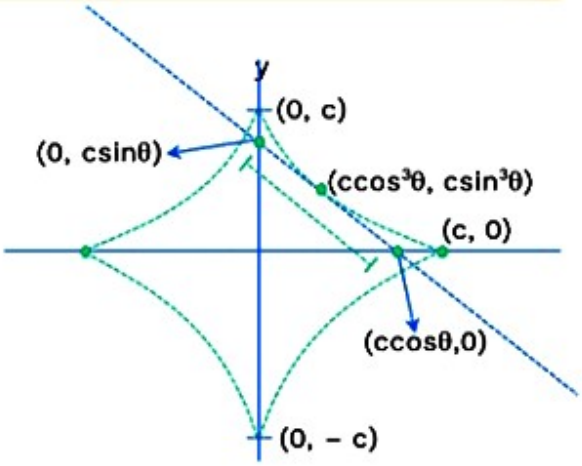
Q.6 **Prove that the portion of the tangent to the curve $x^{2/3} + y^{2/3} = c^{2/3}$ intercepted between the coordinate axes is constant.**

Sol. $\frac{dy}{dx} = \frac{c.3\sin^2\theta\cos\theta}{c.3\cos^2\theta(-\sin\theta)} \Rightarrow \frac{dy}{dx} = -\frac{\sin\theta}{\cos\theta}$

$(y - c\sin^3\theta) = \frac{-\sin\theta}{\cos\theta}(x - c\cos^3\theta)$

$\frac{y}{\sin\theta} - c\sin^2\theta = -\frac{x}{\cos\theta} + c\cos^2\theta$

$\frac{x}{\cos\theta} + \frac{y}{\sin\theta} = c$



Length of required portion

$= \sqrt{(c\cos\theta - 0)^2 + (c\sin\theta - 0)^2}$

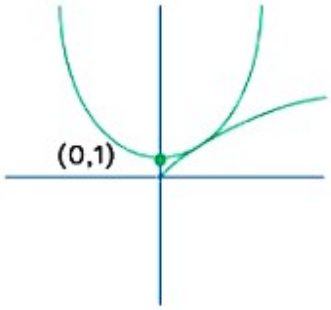
$= \sqrt{c^2\left(\cos^2\theta + \sin^2\theta\right)} = c$

It is constant.

Q.7 **If $y = 1 + \frac{x^2}{a^3}$ and $y = 4\sqrt{x}$ have only one point in common. Find 'a'.**

Sol. $y = 1 + \frac{x^2}{a^3}$ and $y = 4\sqrt{x}$

$a \neq 0$, for $a > 0$



Slope of both curves will be equal to

$\frac{2x}{a^3} = \frac{4}{2\sqrt{x}}$

$x^{3/2} = a^3$

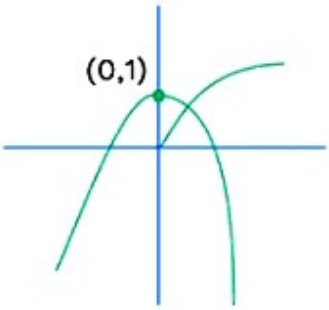
$x = a^2$

by curve, we get

$1 + a = 4a$

$3a = 1 \quad \Rightarrow a = \frac{1}{3}$

for $a < 0$



So when $a < 0$ there will always be a common point

so $a \in (-\infty, 0) \cup \left\{\frac{1}{3}\right\}$

Q.8 **If $y = e^x$ and $y = kx^2$ touches each other. Find 'k'.**

Sol. $y = e^x$, $y = kx^2$

Slope of both the curves at their point of intersection is same.

$e^x = kx^2$

and $e^x = 2kx$

so $kx^2 = 2kx$

$x = 2$

so we get $e^2 = 4k$

$$k = \frac{e^2}{4}$$

Angle of intersection of two curves

It is defined as the angle between the tangents at their point of intersection.

Let (x_1, y_1) be their point of intersection

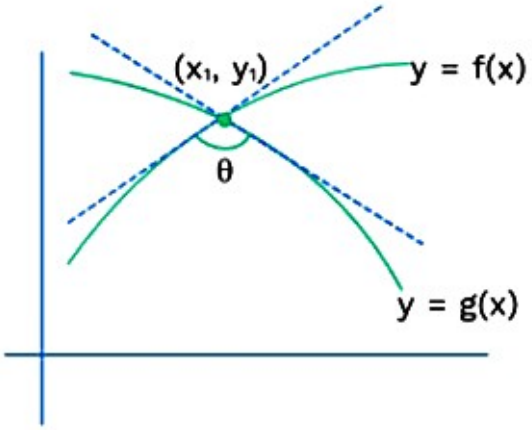
$$m_1 = \left.\frac{df(x)}{dx}\right|_{(x_1,y_1)}$$

$$m_2 = \left.\frac{dg(x)}{dx}\right|_{(x_1,y_1)}$$

$$\boxed{\tan\theta = \left|\frac{m_1 - m_2}{1 + m_2 m_1}\right|}$$, where θ is acute angle.

m_1 and m_2 are finite.

$\pi - \theta$ is the obtuse angle



Note :

If 2 curves intersect each other at 90° at all point of intersections then they are said to be orthogonal.

Mathematically, $m_1.m_2 = -1$

(where m_1 and m_2 are finite)

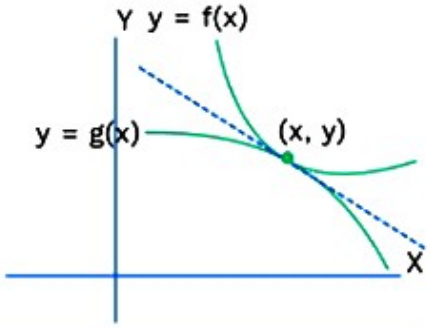
Note : If $m_1 \to \infty$, $m_2 = m$ (finite)

$$\text{Then angle, } \tan\theta = \left|\frac{m_1\left(1 - \frac{m_2}{m_1}\right)}{m_1\left(\frac{1}{m_1} + m_2\right)}\right| = \left|\frac{1}{m_2}\right|$$

- If $m_1 \to \infty$, $m_2 \to \infty$ then angle between them is zero.
- Two curves touch each other, then solve :

  (a) $f(x) = g(x)$

  (b) $f'(x) = g'(x)$



Q.1 **Find the angle between the curves $y = \sin x$ and $y = \cos x$.**

Sol. $\sin x = \cos x$

$\Rightarrow \tan x = 1$

$\Rightarrow x = n\pi + \frac{\pi}{4} \forall n \in I$

For $y = \sin x, y' = m_1 = \frac{1}{\sqrt{2}}$ at $x = \frac{\pi}{4}$

For $y = \cos x$, $y' = m_2 = \frac{-1}{\sqrt{2}}$ at $x = \frac{\pi}{4}$

Angle between the curves $= \tan^{-1}\left(\frac{\frac{1}{\sqrt{2}} - \left(\frac{-1}{\sqrt{2}}\right)}{1 + \frac{1}{\sqrt{2}}\left(\frac{-1}{\sqrt{2}}\right)}\right) = \tan^{-1}\left(\frac{\sqrt{2}}{1 - \frac{1}{2}}\right)$

$= \tan^{-1}\left(2\sqrt{2}\right)$

Q.2 **Find the angle between the curve $2y^2 = x^3$ and $y^2 = 32x$.**

Sol. $2y^2 = x^3$ and $y^2 = 32x$

$\Rightarrow 2(32x) = x^3$

$\Rightarrow 64x - x^3 = 0$

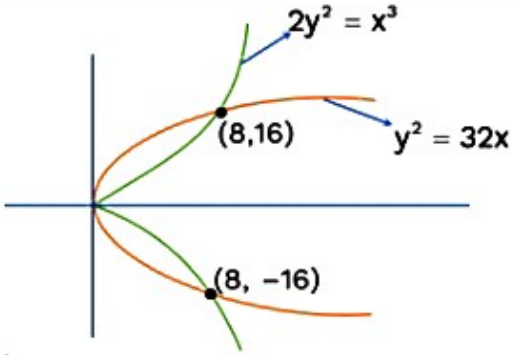
$\Rightarrow x(64 - x^2) = 0$

$\Rightarrow x = 0, 8, -8$.

(where $x = -8$ is not possible)

$\therefore$ Point of intersection are (0, 0), (8, 16), (8,–16)

For $2y^2 = x^3$, differentiating,



$4yy' = 3x^2$

$\Rightarrow y' = \dfrac{3x^2}{4y}$

at $x = 0, y' = 0$ for $2y^2 = x^3$

at $x = 0, y' = \infty$ (vertical tangent) for $y^2 = 32x$.

$\therefore$ angle between curves is $90°$ at (0, 0).

At $(8, 16)$, for $2y^2 = x^3$,

$y' = m_1 = \dfrac{3 \times 8 \times 8}{4 \times 16} = 3$

For $y^2 = 32x$

$y' = m_2 = \dfrac{16}{y} = 1$

$\therefore$ angle between curves $= \tan^{-1}\left(\dfrac{3-1}{1+3}\right) = \tan^{-1}\left(\dfrac{1}{2}\right)$

At $x = (8, -16)$, for $2y^2 = x^3$

$y' = m_1 = -3$

for $y^2 = 32x$

$y' = m_2 = -1$.

$\therefore$ angle between curves $= \tan^{-1}\left(\dfrac{-1+3}{1+3}\right) = \tan^{-1}\left(\dfrac{1}{2}\right)$.

Q.3 **If θ is the angle between $y = x^2$ and $6y = 7 - x^3$ at (a, a), find θ.**

Sol. (a, a) lies on $y = x$

$\therefore y = x = x^2$

$\Rightarrow x = 0, 1$

(0, 0) does not satisfy $6y = 7 - x^3$ while (1, 1) lies on it

$\therefore a = 1$

Now at (1, 1),
for $y = x^2$, $y' = 2x = 2$

for $6y = 7 - x^3$, $y' = \dfrac{-3x^2}{6} = -\dfrac{1}{2}$

$\therefore$ angle between the curves is $90° \left(m_1 m_2 = 2\left(\dfrac{-1}{2}\right) = -1\right)$

Q.4 **Find the condition for the line $y = mx$ to cut at right angles the conic $ax^2 + 2hxy + by^2 = 1$.**

Sol. $y = mx \Rightarrow y' = m_1 = m$

$ax^2 + 2hxy + by^2 = 1$

$2ax + 2hy + 2hxy' + 2byy' = 0$

$$\Rightarrow y' = m_2 = \frac{-(ax + hy)}{hx + by}$$

$$= \frac{-\left(a + \frac{hy}{x}\right)}{h + \frac{by}{x}}$$

$$= \frac{-(a + hm)}{h + bm} \quad (\because y = mx)$$

Since line cuts the curve at right angles , $m_1m_2 = -1$

$$\Rightarrow -m\frac{(a + hm)}{h + bm} = -1$$

$$\Rightarrow m(a + hm) = (h + bm)$$

Q.5 **Show that the condition that the curves**
$x^{\frac{2}{3}} + y^{\frac{2}{3}} = c^{\frac{2}{3}}$ and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \ (a > b > 0)$ **may touch is $c = a + b$.**

Sol. Parametric coordinates of 2 curves are $(c\cos^3\theta, c\sin^3\theta)$ and $(a\cos\alpha, b\sin\alpha)$

$$\Rightarrow c\cos^3\theta = a\cos\alpha$$

$$c\sin^3\theta = b\sin\alpha$$

$$\Rightarrow \tan^3\theta = \frac{b}{a}\tan\alpha \qquad \text{...(i)}$$

Also $\frac{dy_1}{dx} = \frac{dy_2}{dx}$

$$\Rightarrow \frac{3c\sin^2\theta\cos\theta}{-3c\cos^2\theta\sin\theta} = \frac{b\cos\alpha}{-a\sin\alpha}$$

$$\Rightarrow \tan\theta = \frac{b}{a}\cot\alpha \qquad \text{...(ii)}$$

Using (i) and (ii),

$$\left(\frac{b}{a}\right)^3 \cot^3\alpha = \frac{b}{a}\tan\alpha$$

$$\Rightarrow \frac{b^2}{a^2} = \tan^4\alpha \Rightarrow \tan\alpha = \sqrt{\frac{b}{a}}$$

$$\Rightarrow \tan\theta = \frac{b}{a}\sqrt{\frac{a}{b}} = \sqrt{\frac{b}{a}}$$

$$\Rightarrow \tan\alpha = \tan\theta$$

$$\Rightarrow \theta = \alpha$$

$$\Rightarrow c\cos^3\theta = a\cos\theta$$

$$\Rightarrow c\cos^2\theta = a$$

$$\Rightarrow c\sin^2\theta = b$$

Adding both equation,

$c = a + b$ **Hence proved.**

Q.6 **Find the values of a if the curves $\frac{x^2}{a^2} + \frac{y^2}{4} = 1$ and $y^2 = 16x$ cut orthogonally.**

Sol. Differentiating, we get

$$\frac{2x}{a^2} + \frac{2yy'}{4} = 0 \Rightarrow y' = \frac{-4x}{a^2y}$$

And

$$2yy' = 16 \Rightarrow y' = \frac{8}{y}$$

Since both curve cuts orthogonally, $m_1m_2 = -1$

$$\Rightarrow \frac{8}{y}\left(\frac{-4x}{a^2y}\right) = -1$$

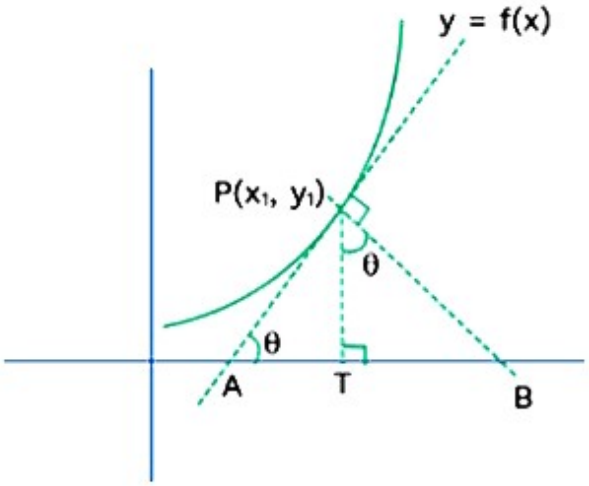
$$\Rightarrow 32x = a^2y^2$$

$$\Rightarrow 32x = a^2(16x)$$

$$\Rightarrow a^2 = 2$$

$$\Rightarrow a = \pm\sqrt{2}$$

Length of tangent, subtangent, normal and subnormal :



Length of tangent at point P is defined as the portion of the tangent intersected between the point of contact and X-axis.

'AP' is the length of tangent.

'AT' is the length of subtangent.

Portion of the normal intercepted between point 'P' and X-axis is length of normal.

'PB' is the length of normal.

'BT' is the length of subnormal.

Numerically :

$$m = \left.\frac{dy}{dx}\right|_{(x_1,y_1)} = \tan\theta$$

In ΔATP, $\quad \sin\theta = \frac{y_1}{PA}$

$$PA = \frac{y_1}{\sin\theta} = y_1 \operatorname{cosec}\theta$$

$$PA = y_1\sqrt{1+\cot^2\theta}$$

$$\text{Length of tangent} = \left| y_1\sqrt{1+\left(\frac{dx}{dy}\right)^2}\right|_{(x_1,y_1)}$$

$$\text{Length of subtangent} = \left|\frac{y_1}{\left.\frac{dy}{dx}\right|_{(x_1,y_1)}}\right| = \left|y_1\left.\frac{dx}{dy}\right|_{(x_1,y_1)}\right|$$

$$\text{Slope of normal} = -\frac{1}{m} = \frac{-1}{\frac{dy}{dx}} = \frac{-dx}{dy}$$

$$\text{Length of normal} = \left| y_1\sqrt{1+\left(\frac{dy}{dx}\right)^2}\right|_{(x_1,y_1)}$$

$$\text{Length of subnormal} = \left|\left. y_1\frac{dy}{dx}\right|_{(x_1,y_1)}\right|$$

Q.1 **Which of the following curves are orthogonal ?**

(A) $y^2 = 4ax$ **and** $y = e^{-\frac{x}{2a}}$

(B) $xy = a^2$ **and** $x^2 - y^2 = b^2$

(C) $y = ax$ **and** $x^2 + y^2 = c^2$

(D) $y^2 = 4ax$ **and** $x^2 = 4by$

Sol. **(A)** At point (x_1, y_1), their slopes are

$2y_1y' = 4a$ and $y' = \frac{-1}{2a}e^{-\frac{x_1}{2a}}$

$y' = \frac{2a}{y_1}$ and $y' = -\frac{1}{2a}y_1$

product of their slopes $= \frac{2a}{y_1}\left(-\frac{y_1}{2a}\right) = -1$

Hence orthogonal.

(B) $xy = a^2$ and $x^2 - y^2 = b^2$

$x_1y' + y_1 = 0$ and $2x_1 - 2y_1y' = 0$

$y' = -\frac{y_1}{x_1}$ and $y' = \frac{x_1}{y_1}$

product of their slopes: $-\frac{y_1}{x_1}.\frac{x_1}{y_1} = -1$

Hence orthogonal.

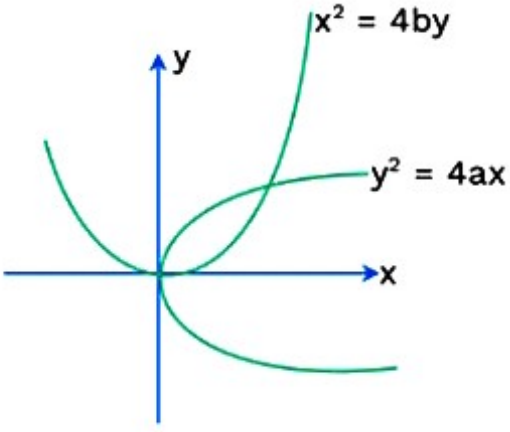
(C) $y = ax$ and $x^2 + y^2 = c^2$

$y' = a$ and $2x_1 + 2y_1y' = 0$

$y' = a = \frac{y_1}{x_1}$ and $y' = -\frac{x_1}{y_1}$

product of their slopes $= \frac{y_1}{x_1}\left(-\frac{x_1}{y_1}\right) = -1$

Hence orthogonal.



(D) $y^2 = 4ax$ and $x^2 = 4by$

When they intersect at (0, 0), they are orthogonal.

Their other point of intersection,

$\frac{x^4}{16b^2} = 4ax$

$\Rightarrow$ $x = 0$ and $x^3 = 64ab^2 \Rightarrow x = 4a^{\frac{1}{3}}b^{\frac{2}{3}}$

$y = 0$ and $y = 4a^{\frac{2}{3}}b^{\frac{1}{3}}$

so at $(4a^{\frac{1}{3}}b^{\frac{2}{3}}m, 4a^{\frac{2}{3}}, b^{\frac{1}{3}})$ their slopes,

$$y' = \frac{2a}{y} = \frac{2a}{4a^{2/3}b^{1/3}} \text{ and } \frac{dy}{dx} = \frac{x}{2b} = \frac{4a^{1/3}b^{2/3}}{2b}$$

$$\frac{dy}{dx} = 2a^{1/3}b^{-1/3}$$

Product of their slopes is not equal to – 1.
So orthogonal at (0, 0) only

Q.2 **Find angle of intersection of the curves $x^3 + y^3 - 3xy = 0$ and $y^2 = x$ at a point other than origin.**

Sol. We have,
$y^6 + y^3 - 3y^3 = 0 \Rightarrow y^6 - 2y^3 = 0$
$\Rightarrow \quad y^3(y^3 - 2) = 0$
$\Rightarrow \quad y = 0, y = 2^{1/3}$
for $y = 2^{1/3} \Rightarrow x = 2^{2/3}$
so point is $(4^{1/3}, 2^{1/3})$ and (0, 0)

Slopes :

For $x^3 + y^3 - 3xy = 0$,
$3x^2 + 3y^2y' - 3xy' - 3y = 0$

$$y' = \frac{y - x^2}{y^2 - x}$$

at $\left(4^{\frac{1}{3}}, 2^{\frac{1}{3}}\right)$, denominator is 0, hence tangent is parallel to y-axis.

and for $y^2 = x$, $2yy' = 1 \Rightarrow y' = \frac{1}{2y} = \frac{1}{2.2^{1/3}} = \frac{1}{16^{1/3}}$

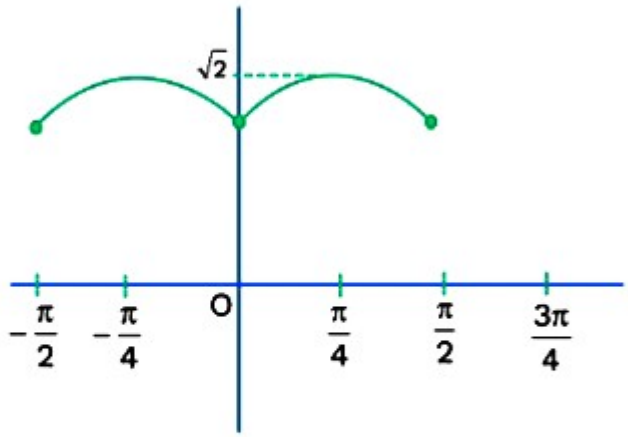
angle between the curves : $\tan\theta = 16^{-1/3}$

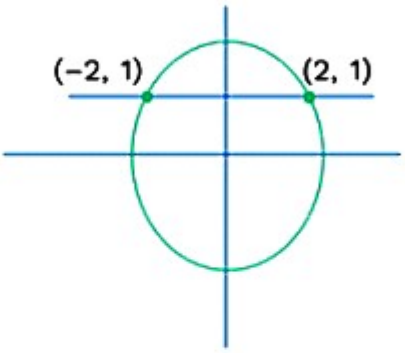
$$\boxed{\theta = \tan^{-1}(16^{1/3})}$$

Q.3 **Find the angle of intersection of the curves $x^2 + y^2 = 5$ and $y = [|\sin x| + |\cos x|]$, where [·] is greatest integer function.**

Sol. $g(x) = |\sin x| + |\cos x|$

$$g(x) = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right) \text{ for } 0 < x < \frac{\pi}{2}$$





g(x) is a periodic function with period π

Hence, $1 \le |\sin x| + |\cos x| \le \sqrt{2}$

and $[|\sin x| + |\cos x|] = 1$

So their point of intersection is (−2, 1) and (2, 1)

Now, slope of the curve $[|\sin x| + |\cos x|]$ is 0

and $2x + 2yy' = 0 \Rightarrow y' = -\frac{x}{y}$ is slope for circle

$y'|_{(2,1)} = -2$ and $y'|_{(-2,1)} = 2$

hence $\tan\theta = 2 \Rightarrow \boxed{\theta = \tan^{-1} 2}$

Q.4 **Find the condition for which the two curves $a_1x^2 + b_1y^2 = 1$ and $a_2x^2 + b_2y^2 = 1$ intersect orthogonally.**

Sol. Suppose their point of intersection is (x_1, y_1)

then $2a_1x_1 + 2b_1y_1y' = 0 \Rightarrow y' = \frac{-a_1x_1}{b_1y_1}$

and $2a_2x_1 + 2b_2y_1y' = 0 \Rightarrow y' = \frac{-a_2x_1}{b_2y_1}$

product of their slopes = − 1

So we get, $-\frac{a_1x_1}{b_1y_1}.\frac{-a_2x_1}{b_2y_1} = -1$

$$\frac{a_1a_2x_1^2}{b_2b_1y_1^2} = -1 \qquad \ldots(i)$$

also $a_1x_1^2 + b_1y_1^2 = 1$

$a_2x_1^2 + b_2y_1^2 = 1$

by subtraction,

$(a_1 - a_2)x_1^2 + (b_1 - b_2)y_1^2 = 0$

$$\frac{x_1^2}{y_1^2} = -\frac{(b_1 - b_2)}{(a_1 - a_2)} \qquad \ldots(ii)$$

By (i) and (ii),

$$\frac{a_1 a_2}{b_1 b_2}\left[\frac{-(b_1 - b_2)}{(a_1 - a_2)}\right] = -1$$

$$\Rightarrow \quad \frac{b_1 - b_2}{b_1 b_2} = \frac{a_1 - a_2}{a_1 a_2}$$

$$\Rightarrow \quad \frac{1}{b_2} - \frac{1}{b_1} = \frac{1}{a_2} - \frac{1}{a_1}$$

$$\Rightarrow \quad \boxed{\frac{1}{a_2} + \frac{1}{b_1} = \frac{1}{a_1} + \frac{1}{b_2}}$$

Q.5 **Length of the normal to the curve** $x = a(\theta + \sin\theta), y = a(1 - \cos\theta)$**, at** $\theta = \frac{\pi}{2}$ **is**

(A) $2a$ **(B)** $a\sqrt{2}$ **(C)** $\frac{a}{2}$ **(D)** $\frac{a}{\sqrt{2}}$

Sol. **(B)**

At $\theta = \frac{\pi}{2}, \frac{dy}{dx} = \frac{\sin\theta}{1 + \cos\theta} = 1 = \tan\alpha$

Length of normal $= |y \sec\alpha| = a\sqrt{2}$ $\left(\because y = a \text{ at } \theta = \frac{\pi}{2}\right)$

Q.6 **Find the acute angle between the curves $y = |x^2 - 1|$ and $y = |x^2 - 3|$ at their points of intersection.**

Sol. Points of intersection:

$x^2 - 1 = x^2 - 3$ (No solution)

Or

$x^2 - 1 = 3 - x^2$

$\Rightarrow x^2 = 2 \Rightarrow x = \sqrt{2}, -\sqrt{2}$

$\Rightarrow y_1 = x^2 - 1 \quad ; \quad y_2 = 3 - x^2$

$\Rightarrow y_1' = m_1 = 2x \quad ; y_2' = m_2 = -2x$

$= 2\sqrt{2} \qquad = -2\sqrt{2} \qquad (\text{at } x = \sqrt{2})$

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

$$= \left|\frac{2\sqrt{2} - \left(-2\sqrt{2}\right)}{1 + \left(2\sqrt{2}\right)\left(-2\sqrt{2}\right)}\right|$$

$$\tan\theta = \frac{4\sqrt{2}}{7} \Rightarrow \theta = \tan^{-1}\left(\frac{4\sqrt{2}}{7}\right)$$

Q.7

The length of the normal to the curve $y = a\dfrac{\left(e^{\frac{-x}{a}} + e^{\frac{x}{a}}\right)}{2}$ **at any point varies as the**

(A) abscissa of the point
(B) ordinate of the point
(C) square of the abscissa of the point
(D) Square of the ordinate of the point.

Sol. **(D)**

Length of Normal $= |y\sec\theta|$ where $\tan\theta = \dfrac{dy}{dx}$

$$\therefore \frac{dy}{dx} = \frac{a}{2}\left(\frac{-e^{\frac{-x}{a}}}{a} + \frac{e^{\frac{x}{a}}}{a}\right)$$

$$= \left(\frac{e^{\frac{x}{a}} - \frac{1}{e^{x/a}}}{2}\right)$$

Let $e^{\frac{x}{a}} = \alpha$

$\Rightarrow$ Length of Normal $= |y\sec\theta|$

$$= \left|y\sqrt{1 + \tan^2\theta}\right| = \left|\frac{a}{2}\left(\alpha + \frac{1}{\alpha}\right)\sqrt{1 + \left(\frac{\alpha - \frac{1}{\alpha}}{2}\right)^2}\right|$$

$$= \left| \frac{a}{2}\left(\alpha + \frac{1}{\alpha}\right)\sqrt{\left(\frac{\alpha + \frac{1}{\alpha}}{2}\right)^2} \right| = \frac{a}{4}\left(\alpha + \frac{1}{\alpha}\right)^2 = \frac{y^2}{a}$$

So, length of normal varies as the square of ordinate of the point.

Q.8 **The subtangent at any point on the curve $x^m y^n = a^{m+n}$ varies as**
(A) (abscissa)^2 (B) (ordinate)^2 (C) abscissa (D) Ordinate

Sol. **(C)**

Length of subtangent $= \left| y\frac{dx}{dy} \right|$

$x^m y^n = a^{m+n}$

On differentiating, we get

$mx^{m-1}y^n + nx^m y^{n-1}y' = 0$

$\Rightarrow my + nxy' = 0$

$\Rightarrow y' = \frac{-my}{nx}$

$\therefore$ Length of subtangent $= \left| \frac{y}{\left(\frac{-my}{nx}\right)} \right| = \left| \frac{nx}{m} \right|$

Q.9 **Determine p such that the length of the sub-tangent and sub-normal is equal for the curve $y = e^{px} + px$ at the point (0, 1).**

Sol. Length of subtangent $= \left| y\frac{dx}{dy} \right|$

Length of subnormal $= \left| y\frac{dy}{dx} \right|$

$\Rightarrow \left| y\frac{dx}{dy} \right| = \left| y\frac{dy}{dx} \right|$

$\Rightarrow \left(\frac{dy}{dx}\right)^2 = 1 \Rightarrow \frac{dy}{dx} = \pm 1$

$\Rightarrow pe^{px} + p = 1 \text{ or } -1 \text{ at } x = 0$

$\Rightarrow 2p = 1 \text{ or } -1$

$\Rightarrow p = \frac{1}{2} \text{ or } \frac{-1}{2}$

Q.10 **Find the value of k so that the length of the subnormal at any point on the curve $y = a^{1-k} x^k$ is a constant.**

Sol. $\left|y\frac{dy}{dx}\right|$ = length of subnormal = constant

$\frac{dy}{dx} = ka^{1-k}x^{k-1}$

$y\frac{dy}{dx} = k\left(a^{1-k}\right)^2 x^{2k-1}$ = constant

$\Rightarrow 2k - 1 = 0 \Rightarrow k = \frac{1}{2}.$

Q.11 **Prove that for the curve $by^2 = (x + a)^3$, the square of the subtangent varies as the subnormal**

Sol. $(S.T.)^2 \propto (S.N.)$
$(S.T.)^2 = k(S.N.)$

$\frac{(S.T.)^2}{S.N.} = k$

Slope :

$2bym = 3(x + a)^2 \Rightarrow m = \frac{3(x+a)^2}{2by}$

Subtangent $= \left|\frac{y.2by}{3(x+a)^2}\right|$ and subnormal $= \left|\frac{3(x+a)^2 y}{2by}\right|$

Now $\frac{(S.T.)^2}{(S.N.)} = \left|\frac{4b^2y^4 \quad 2b}{9(x+a)^4.3(x+a)^2}\right|$

$\frac{(S.T.)^2}{(S.N.)} = \left|\frac{8b^3y^4}{27(x+a)^6}\right| = \left|\frac{8b \quad b^2y^4}{27(x+a)^6}\right|$

$\frac{(S.T.)^2}{(S.N.)} = \left|\frac{8b}{27}\right|$ = constant **Hence proved.**

Rate Measure

Functions will be differentiated with respect to variable time 't'.

Q.1 **If the area of circle increases at a uniform rate, prove that the rate of increase of the perimeter varies inversely to its radius.**

Sol. $A = \pi r^2$

$$\frac{dA}{dt} = k \Rightarrow \pi 2r\frac{dr}{dt} = k \Rightarrow \frac{dr}{dt} = \frac{k}{2\pi r}$$

$P = 2\pi r$

$$\frac{dP}{dt} = 2\pi\frac{dr}{dt} = \frac{2\pi k}{2\pi r} = \frac{k}{r}$$

$$\boxed{\frac{dP}{dt} \propto \frac{1}{r}}$$

Hence proved.

Q.2 **If the side of an equilateral triangle increase at a rate of $\sqrt{3}$ cm./sec. and area increase at the rate of 12 cm².sec, then the side of equilateral triangle is**

(A) 2 **(B) 4** **(C) 6** **(D) 8**

Ans. **(D)**

Sol. $$A = \frac{\sqrt{3}}{4}a^2$$

$$\frac{dA}{dt} = \frac{\sqrt{3}}{4}2a\frac{da}{dt}$$

$$\Rightarrow 12 = \frac{\sqrt{3}}{4}2a\sqrt{3} \Rightarrow a = 8$$

Q.3 **The height of a triangle circular cone is 20 cm and is decreasing at the rate of 4cm/sec. At the same time, the radius is 10 cm and is increasing at the rate of 2 cm/sec. Find the rate of change of volume in cm³/sec.**

(A) $\frac{100\pi}{3}$ **(B)** $\frac{200\pi}{3}$ **(C)** $\frac{400\pi}{3}$ **(D)** $\frac{800\pi}{3}$

Sol. **(C)**

$h = 20$ $\quad$ $r = 10$

$$\frac{dh}{dt} = -4 \qquad \frac{dr}{dt} = 2$$

$$V = \frac{\pi}{3} r^2 h$$

$$\frac{dV}{dt} = \frac{\pi}{3}\left(h.2r\frac{dr}{dt} + r^2\frac{dh}{dt}\right)$$

$$\frac{dV}{dt} = \frac{\pi}{3}[20.20.2 - 100 \times 4]$$

$$\frac{dV}{dt} = \frac{400\pi}{3}$$

Q.4 **If the radius of a circle is increasing at a uniform rate of 2cm/s, find the rate of increase of area of circle, at the instant when radius is 20 cm.**

Sol. $\frac{dr}{dt} = 2cm/s$

$r = 20cm$

$A = \pi r^2$

$$\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$$

$$= 2\pi(20)2$$

$$= 80\pi cm^2/sec$$

Q.5 **If $s = \frac{1}{2}t^3 - 6t$, find the acceleration at time when the velocity is zero. (s is displacement).**

Sol. $$V = \frac{ds}{dt} = \frac{3t^2}{2} - 6$$

$$\Rightarrow v = 0 = \frac{3t^2}{2} - 6$$

$\Rightarrow t^2 = 4$

$\Rightarrow t = 2sec.$

$$a = \frac{dv}{dt} = 3t = 6.$$

Q.6 **If r be the radius, S be the surface area and V be the volume of a spherical bubble, prove that**

(i) $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$ **(ii)** $\frac{dV}{ds} \propto r$

Sol. **(i)** $V = \frac{4}{3}\pi r^3$

$$\Rightarrow \frac{dV}{dt} = \frac{4}{3}\pi\left(3r^2\right)\frac{dr}{dt}$$

$$= 4\pi r^2 \frac{dr}{dt}$$

(ii) $S = 4\pi r^2$

$$\Rightarrow \frac{dS}{dt} = 8\pi r \frac{dr}{dt}$$

$$\frac{dV}{dS} = \frac{\frac{dV}{dt}}{\frac{dS}{dt}} = \frac{4\pi r^2 \frac{dr}{dt}}{8\pi r \frac{dr}{dt}}$$

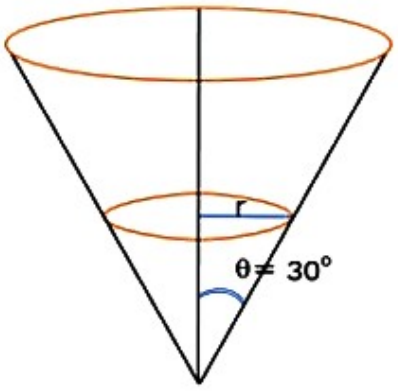
$$\Rightarrow \frac{dV}{dS} = \frac{r}{2}$$

$$\Rightarrow \frac{dV}{dS} \propto r$$

Hence proved.

Q.7 **If water is poured into an inverted hollow cone whose semi-vertical angle is 30°, its depth (measured along the axis) increases at the rate of 1 cm/sec. Find the rate at which the volume of water increases when the depth is 24 cm.**

Sol. $\frac{dh}{dt} = 1, h = 24$



$$\tan 30° = \frac{1}{\sqrt{3}} = \frac{r}{h}$$

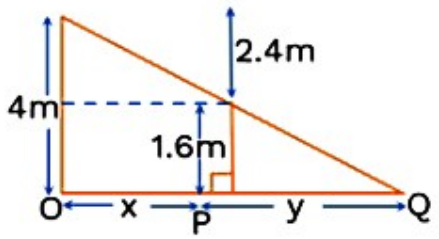
$$\Rightarrow r = \frac{h}{\sqrt{3}} = 8\sqrt{3}$$

$$V = \frac{1}{3}\pi r^2 h = \frac{\pi h^3}{9}$$

$$\frac{dV}{dt} = \frac{3\pi h^2}{9}\frac{dh}{dt} = \frac{\pi(24)(24)}{3}(1) = 192\pi \text{ cm}^3/\text{s}$$

Q.8 **A man who is 1.6 m tall walks away from a lamp which is 4 m above the ground at the rate of 30 m/min. How fast is the man's shadow lengthening?**

Sol.



$$\frac{2.4}{x} = \frac{1.6}{y}$$

$$\Rightarrow y = \frac{2x}{3}$$

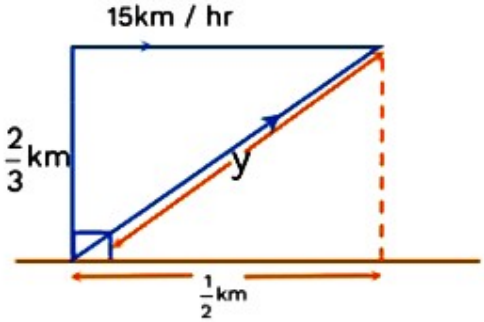
$$\Rightarrow \frac{dy}{dt} = \frac{2}{3}\frac{dx}{dt}$$

$$\frac{dx}{dt} = 30\text{m/min}$$

Rate of change of shadow $= \frac{dy}{dt} = \frac{2}{3} \times 30 = 20\text{m/min}$

Q.9 **An aeroplane is flying horizontally at a height of $\frac{2}{3}$km with a velocity of 15 km/ hr. Find the rate at which it is receding from a fixed point on the ground which it passed over 2 minutes ago.**

Sol. Let x be the distance travelled by plane.



$$\Rightarrow y^2 = x^2 + \left(\frac{2}{3}\right)^2$$

$$\Rightarrow 2y\frac{dy}{dt} = 2x\frac{dx}{dt}$$

$$\Rightarrow \frac{dy}{dt} = \frac{x}{y}\frac{dx}{dt}$$

After 2 minute, $x = \frac{1}{2}$km

$$y = \sqrt{\left(\frac{2}{3}\right)^2 + \left(\frac{1}{2}\right)^2} = \frac{5}{6}\text{km}$$

Also $\frac{dx}{dt} = 15\text{km/hr} \Rightarrow \frac{dy}{dt} = \frac{\frac{1}{2}}{\frac{5}{6}}(15)$

$$\Rightarrow \frac{dy}{dt} = 9\text{km/hr}$$

Q.10 **The radius of the base of a cone is increasing at the rate of 3cm/minute and the altitude is decreasing at the rate of 4cm/minute. The rate of change of lateral surface area when the radius = 7cm and the altitude = 24 cm is**

(A) $54\pi\,cm^2$ / min **(B)** $7\pi\,cm^2$ / min

(C) $27\,cm^2$ / min **(D) None of these**

Sol. **(A)**

$S = \pi r\ell$

$$\frac{dS}{dt} = \frac{\pi r dl}{dt} + \pi\ell\frac{dr}{dt}$$

$$\frac{dr}{dt} = 3,\ \frac{dh}{dt} = 4$$

$$\ell = \sqrt{r^2 + h^2}$$

$$\frac{d\ell}{dt} = \frac{1}{2\sqrt{r^2+h^2}}\left(2r\frac{dr}{dt} + \frac{2hdh}{dt}\right)$$

At $r = 7, h = 24, \ell = 25$

$$\Rightarrow \frac{d\ell}{dt} = \frac{1}{2\times 25}\left(2\times 7\times 3 + 2\times 24\times(-4)\right)$$

$$= \frac{1}{25}(21 - 96) = -3$$

$$\Rightarrow \frac{dS}{dt} = \pi(7)(-3) + \pi(25)(3) = 54\pi\ cm^2\ /\ min$$

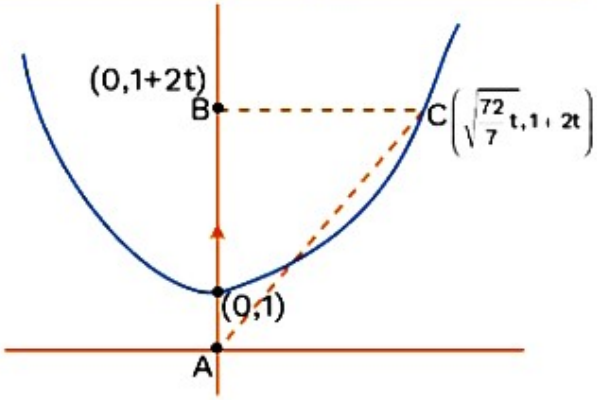
Q.11 **A variable ΔABC in the x-y plane has its orthocentre at vertex 'B', a fixed 'A' at the origin and the third vertex 'C' restricted to lie on the parabola $y = 1 + \frac{7x^2}{36}$. The point B starts at the point (0, 1) at time t = 0 and moves upwards of 2cm/sec. How fast is the area of the triangle increasing when $t = \frac{7}{2}$ sec?**

Sol.

$$\text{Area} = \frac{1}{2}(1+2t)\sqrt{\frac{72}{7}}t$$

$$A = \sqrt{\frac{18}{7}}\left(t^{1/2} + 2t^{3/2}\right)$$

$$\frac{dA}{dt} = \sqrt{\frac{18}{7}}\left(\frac{1}{2\sqrt{t}} + 3t^{\frac{1}{2}}\right)$$



$\therefore$ at $t = \frac{7}{2}$,

$$\frac{dA}{dt} = \sqrt{\frac{18}{7}}\left(\frac{1}{2}\sqrt{\frac{2}{7}} + 3\sqrt{\frac{7}{2}}\right)$$

$$= \frac{3}{7} + 9 = \frac{66}{7}\text{cm}^2/\text{sec}$$

Q.12 **Water is flowing out at the rate of $6m^3$/min from a reservoir shaped like a hemispherical bowl of radius R = 13 m. The volume of water in the hemispherical bowl is given by $V = \frac{\pi}{3}y^2(3R - y)$ when the water is 'y' meter deep. Find**

(a) At what rate is the water level changing when the water is 8m deep.

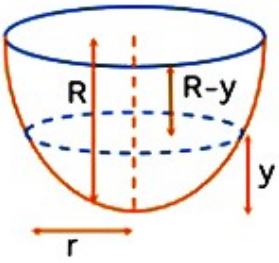
(b) At what rate is the radius of water surface changing when the water is 8 m deep.

Sol. **(a)** $\frac{dV}{dt} = \frac{\pi}{3}\left\{(3R)\left(2y\frac{dy}{dt}\right) - 3y^2\frac{dy}{dt}\right\}$

$$\Rightarrow -6 = \pi\left((13)(2)8\frac{dy}{dt} - (8^2)\frac{dy}{dt}\right)$$

$$\Rightarrow -6 = \pi \times 8 \times 18\frac{dy}{dt}$$

$$\Rightarrow \frac{dy}{dt} = \frac{-1}{24\pi}\text{m/min}$$



(b) $R^2 = r^2 + (R - y)^2$

$$\Rightarrow r^2 = 2Ry - y^2$$

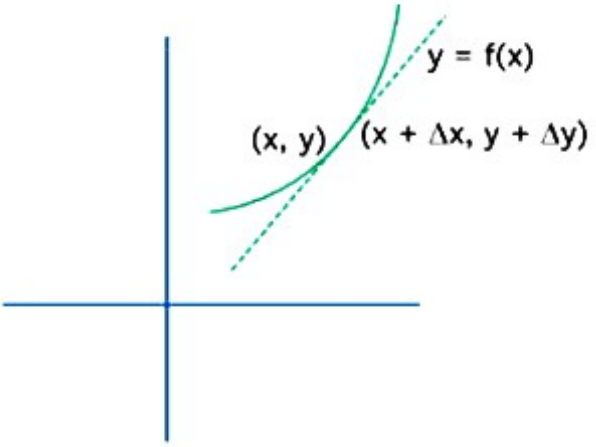
$$\Rightarrow \frac{2rdr}{dt} = \frac{2Rdy}{dt} - \frac{2ydy}{dt}$$

When y = 8, r = 12

$$\Rightarrow 2 \times 12\frac{dr}{dt} = 2 \times 13\frac{dy}{dt} - 2 \times 8\frac{dy}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{5}{12}\left(\frac{-1}{24\pi}\right) = \frac{-5}{288\pi}\text{m/min}$$

Approximation :



Slope = $\frac{\Delta y}{\Delta x}$

when $\Delta x \to 0$

$$\boxed{\frac{\Delta y}{\Delta x} \simeq f'(x)} \qquad ...(i)$$

$y = f(x)$

$y + \Delta y = f(x + \Delta x)$

Subtraction gives,

$\Delta y = f(x + \Delta x) - f(x)$

By (i), $\frac{\Delta y}{\Delta x} \simeq f'(x) \Rightarrow f(x + \Delta x) - f(x) \simeq \Delta x.\ f'(x)$

$$\boxed{f(x + \Delta x) = f(x) + \Delta x.f'(x)}$$

Q.13 **Find the approximate value of $\sqrt{101}$.**

Sol. Suppose $y = f(x) = \sqrt{x}$

$f(101) = \sqrt{101} = \sqrt{100 + 1}$

$f(100 + 1) = \sqrt{100 + 1}$

$x = 100,\ \Delta x = 1$

and $f'(x) = \frac{1}{2\sqrt{x}}$

then $f(101) = f(100) + 1 \cdot \left.\frac{1}{2\sqrt{x}}\right|_{x=100}$

$= 10 + \frac{1}{2\sqrt{100}} = 10 + \frac{1}{20}$

$f(101) = 10.05$

Q.14 **Find the approximate value of $(25)^{\frac{1}{3}}$.**

Sol. Let $y = x^{\frac{1}{3}}$

$x = 27 \Rightarrow y = 3$

$\Delta x = -2$ (for given question)

$$\frac{\Delta y}{\Delta x} \approx \frac{1}{3x^{\frac{2}{3}}} = \frac{1}{3\times 9} = \frac{1}{27}$$

$$\Delta y \approx \frac{\Delta x}{27} = \frac{-2}{27}$$

$$y + \Delta y = 3 - \frac{2}{27} = \frac{79}{27} \qquad \therefore 25^{\frac{1}{3}} \approx \frac{79}{27}$$

Q.15 **Find the approximate value of f(3.02) where $f(x) = 3x^2 + 5x + 3$.**

Sol. At $x = 3$, $y = 27 + 15 + 3 = 45$

$f'(x) = 6x + 5$

$$\Rightarrow \frac{\Delta y}{\Delta x} \approx 6x + 5$$

Here $\Delta x = 0.02$

$$\Rightarrow \Delta y \approx (6\times 3 + 5)(0.02)$$

$$\Rightarrow \Delta y \approx 23(0.02)$$

$$\Rightarrow \Delta y \approx 0.46$$

$$\Rightarrow f(3.02) \approx y + \Delta y \approx 45.46$$

Differential

$y = f(x)$

$$\frac{dy}{dx} = f'(x)$$

It is a composite structure which is used to make slope of the tangent at some point (x, y).

Differential of y i.e. 'dy' is defined as product of derivative of f(x) w.r.t. x and differential of x i.e. dx

$$dy = f'(x)dx$$

Q.1 **Find the differential of $y = \sin x$**

Sol. $dy = f'(x)\,dx$

$dy = \cos x\,dx$

$d(\sin x) = \cos x\,dx$

Q.2 **Find the differential of $y = \tan^{-1}u$**

Sol. $dy = f'(u)\,du$

$$d(\tan^{-1}u) = \frac{1}{1+u^2}du$$

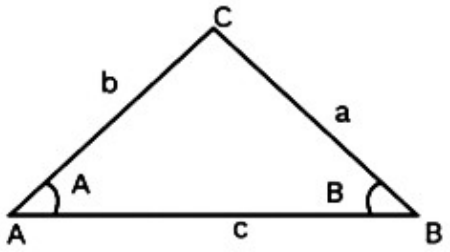
Q.3 **Find the differential of $y = \sin^{-1}x$**

Sol. $dy = f'(x)\,dx$

$$d(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}dx$$

Q.4 **In an acute triangle ABC if sides a, b be constants and base angles A and B vary, then prove that** $\dfrac{dA}{\sqrt{a^2 - b^2\sin^2 A}} = \dfrac{dB}{\sqrt{b^2 - a^2\sin^2 B}}$

Sol. Using sine rule, $\dfrac{a}{\sin A} = \dfrac{b}{\sin B}$



$a\sin B = b\sin A$

$d(a\sin B) = d(b\sin A)$

$a\cos B\,dB = b\cos A\,dA$

$$\frac{dA}{a\cos B} = \frac{dB}{b\cos A}$$

$$\Rightarrow \frac{dA}{\sqrt{a^2(1-\sin^2 B)}} = \frac{dB}{\sqrt{b^2(1-\sin^2 A)}}$$

$$\Rightarrow \frac{dA}{\sqrt{a^2 - b^2\sin^2 A}} = \frac{dB}{\sqrt{b^2 - a^2\sin^2 B}}$$

Hence proved.

Q.5 **If in a triangle ABC, the side C and the angle C remain constant, while the remaining elements are changed slightly then $\frac{da}{\cos A}+\frac{db}{\cos B}$ is equal to**

(A) 0 **(B) 1** **(C) –1** **(D) None of these**

Sol. **(A)**

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{C}{\sin B}=2R=\text{constant}$$

$\Rightarrow a=2R\sin A$ and $b=2R\sin B$

$\Rightarrow da=2R\cos A\,dA$ and $db=2R\cos B\,dB$

$$\Rightarrow \frac{da}{\cos A}+\frac{db}{\cos B}=2R(dA+dB)$$

Now $A+B+C=\pi$

$\Rightarrow A+B=\pi-C$

$\Rightarrow dA+dB=0$ (C is constant)

$$\Rightarrow \frac{dA}{\cos A}+\frac{dB}{\cos B}=0$$

Q.6 **If a particle starts moving from (0, 0) and rate of change of its abscissa is 8 cm/s on the curve $y=\sqrt{x}$, then the rate of change of area of triangle formed by coordinate axes and tangent at a point reached after 2 sec. of start is**

(A) 3 **(B) 6** **(C) 9** **(D) 12**

Sol. $\frac{d\alpha}{dt}=8$, slope $=\frac{1}{2\sqrt{x}}$

$$(y-\sqrt{\alpha})=\frac{1}{2\sqrt{\alpha}}(x-\alpha)$$

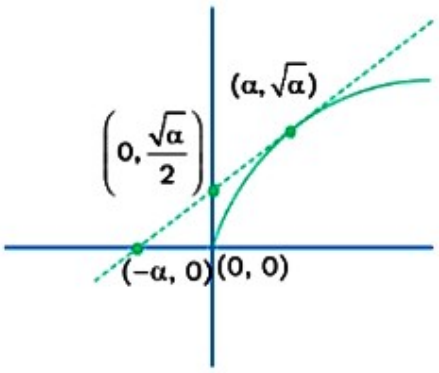
put $x=0\Rightarrow y=\frac{\sqrt{\alpha}}{2}$

$y=0\Rightarrow x=-\alpha$

$$\text{Area}=\frac{1}{2}\alpha.\frac{\sqrt{\alpha}}{2}=\frac{1}{4}\alpha^{3/2}$$

$$\frac{dA}{dt}=\frac{1}{4}\frac{3}{2}\alpha^{1/2}\frac{d\alpha}{dt}$$

$$\frac{dA}{dt}=\frac{1}{4}.\frac{3}{2}.4.8=12 \quad (\because \alpha=16) \qquad \Rightarrow \boxed{\frac{dA}{dt}=12}$$



Question on errors measurement :

Q.1 **If there is 2% error in measuring the radius of sphere, then ... will be the percentage error in the surface area.**

(A) 3% **(B) 1%** **(C) 4%** **(D) 2%**

Sol. **(C)**

Given $\frac{\delta r}{r} \times 100 = 2 \Rightarrow \delta r = \frac{2r}{100}$

Surface area, $S = 4\pi r^2$

$\delta S = 4\pi 2r.\delta r$

$8\pi r.\frac{2r}{100} = \frac{16\pi r^2}{100}$

Now percentage error in surface area

$\frac{\delta S}{s} \times 100 = \frac{16\pi r^2}{100} \times \frac{1}{4\pi r^2} \times 100 = 4\%$

Q.2 **If there is an error of a% in measuring the edge of a cube, then percentage error in its surface is**

(A) 2a% **(B)** $\frac{a}{2}\%$ **(C) 3a%** **(D) None of these**

Sol. **(A)**

Let x be the length of an edge of a cube and S be its surface area. Then,

$S = 6x^2 \Rightarrow \frac{dS}{dx} = 12x$

$\therefore \Delta S = \frac{dS}{dx}\Delta x$

$\Rightarrow \Delta S = 12x\,\Delta x$

$\Rightarrow \frac{\Delta S}{S} \times 100 = \frac{12x\ \Delta x}{6x^2} \times 100 = 2\left(\frac{\Delta x}{x} \times 100\right)$

$= 2a$